

Kinematic Anchor Guidance Enables Zero-Strain Hydration-Site Targeting in 3D Molecular Generation

Author Name(s)

Institution(s)

{corresponding.author}@institution.edu

Abstract

A grand challenge in structure-based 3D molecular generation is achieving *zero-strain* targeting of specific hydration sites: attracting ligand atoms to displace high-energy water molecules (HEWs, worth 0.5–2.0 kcal/mol in binding free energy) without corrupting the internal conformational quality that the generative model has learned from protein–ligand co-crystals. Existing generators suffer from **water-blindness**—their training data lack explicit water molecules, so they never learn to exploit HEW sites—while existing inference-time guidance methods operate only on global molecular properties, incapable of the site-type-level, topology-level control that water displacement demands.

A natural intuition would be to enforce site occupancy by rigidly fixing anchor-atom coordinates (“hard-fix”). We show, via the Kinetic Path Energy (KPE) diagnostic framework, that this intuitively appealing strategy is physically catastrophic: each coordinate overwrite constitutes an instantaneous teleport in the flow-matching ODE, injecting infinite-velocity spikes that cause a **31,468-fold surge** in destructive kinetic energy (98.5% of total ODE energy). This excess energy locks and tears internal conformational degrees of freedom, producing severely strained molecules with degraded Vina docking scores.

We propose **Kinematic Anchor Guidance**, a principled alternative rooted in classical mechanics. The key insight is that hydration-site attraction is fundamentally a centre-of-mass (CoM) operation: the guidance displacement field is orthogonally decomposed into internal conformational ($\mathbf{v}_{\text{int}}$) and rigid-body translational ($\mathbf{v}_{\text{CoM}}$) components, and guiding forces act *exclusively* on the CoM subspace ($\mathbb{R}^3$). This mathematically guarantees **zero strain** on all bond lengths, angles, and torsions—the molecule is softly pulled toward the target site as a rigid body, preserving full internal flexibility for natural conformational relaxation.

Across 6 pharmacologically diverse PDBbind pockets on DrugFlow (3,200+ generated molecules with full Vina docking), plus cross-validation on TargetDiff (a structurally distinct DDPM-based diffusion model), kinematic anchoring delivers: (i) **31,468 $\times$ KPE suppression** ($\rho_{\text{KPE}} = 0.006\%$ vs. 98.5% for hard-fix), providing ironclad physical validation of the zero-strain guarantee; (ii) **simultaneous improvement in both occupancy and affinity**—the best Vina score in every pocket with Wilcoxon signed-rank $p = 4 \times 10^{-6}$ over hard-fix; and (iii) **2 $\times$ variance reduction** ($\sigma_{\text{Vina}}: 0.52 \rightarrow 0.27$), eliminating the catastrophic conformational collapse (Vina = -1.70 kcal/mol) and repulsive outliers ($+6.86$ kcal/mol) that plague hard-fix and annealing baselines. These results establish that physically consistent, zero-strain hydration-site targeting is both achievable and essential for the next generation of water-aware molecular design.

1 Introduction

1.1 Water as a First-Class Design Constraint

Water molecules at protein–ligand interfaces are not passive bystanders. Their thermodynamic signatures—whether a water is “happy” (structurally conserved, mediating critical H-bond networks) or “unhappy” (high-energy, trapped in a hydrophobic sub-pocket)—can be worth 0.5–2.0 kcal/mol per molecule in binding free energy [1, 2, 3]. Displacing a single high-energy water (HEW) with a compatible ligand functional group can therefore contribute binding affinity comparable to a strong hydrogen bond [4, 5].

Computational tools for predicting hydration thermodynamics are well-established: WaterMap [2], Grid Inhomogeneous Solvation Theory (GIST) [6], Grand Canonical Monte Carlo (GCMC) [7], and 3D-RISM [8] each quantify water energetics from molecular simulations or integral equation theory. Yet these tools produce *analyses*—heat maps

and free energy estimates—that medicinal chemists must manually interpret. They have not been integrated as explicit, differentiable design constraints within *de novo* 3D molecular generation pipelines.

1.2 The Water-Blindness of Deep Generative Models

Recent structure-based 3D molecular generators—TargetDiff [9], DiffSBDD [10], DecompDiff [11], DrugFlow [12], and FLAG [13]—have demonstrated impressive capabilities: they generate drug-like molecules directly conditioned on protein pocket geometry, achieving high validity rates and competitive docking scores. At their core, these models learn $p(\text{ligand} \mid \text{pocket})$, a distribution over atom types and coordinates conditioned on protein atoms.

A critical limitation lurks in the training data. Crystallographic structures from the Protein Data Bank [14] that comprise the training corpora rarely retain explicit water molecules at the binding interface. Consequently, the learned distribution $p(\text{ligand} \mid \text{pocket})$ encodes no information about *which* pocket regions should be occupied by ligand atoms versus water molecules. The model has no incentive to displace a HEW—it may equally well place a methyl group adjacent to the water site, leaving the energetically unfavorable water in place and forfeiting a significant binding affinity opportunity. We term this deficiency **water-blindness**.

Our earlier diagnostic work [15, 16] established the severity of this blindness. Across 10 pharmacologically diverse PDBbind pockets, a state-of-the-art DrugFlow generator achieved **0% HEW site occupancy** in unguided mode—not a single generated molecule placed any atom within 2.5 Å of a HEW site. This failure is not due to poor molecular quality: the baseline molecules were chemically valid, drug-like ($\text{QED} = 0.39 \pm 0.14$), and achieved reasonable Vina scores. Rather, the generator’s implicit geometric prior fundamentally diverges from the explicit physicochemical objective of water displacement.

1.3 The Naive Fix and Its Hidden Catastrophe: A KPE Diagnosis

Faced with water-blindness, the most instinctively appealing strategy is straightforward: during generation, *identify atoms near HEW sites, and rigidly fix their coordinates to enforce occupancy*. This “hard-fix” approach—deterministically overwriting selected atom coordinates to target values after each ODE integration step—has been employed in various forms by fragment-fixing strategies in conditional generation [17] and by the two-stage topology-aware framework of ESFIELD v7.1 [15]. The intuition is compelling: if you want atoms at specific locations, simply force them there. What could go wrong?

To answer this question rigorously, we turn to the **Kinetic Path Energy (KPE)** diagnostic framework recently formalized by Li et al. [18]. KPE quantifies the transport cost of a flow-matching trajectory by measuring the integrated squared norm of the effective velocity field:

$$E_{\text{KPE}} = \int_0^1 \mathbb{E}_{\mathbf{x}_t \sim p_t} [\|\mathbf{v}_\theta(\mathbf{x}_t, t)\|^2] dt. \quad (1)$$

Crucially, KPE decomposes into a **physiological component** (the minimal kinetic energy of ODE transport following the learned data manifold) and a **guidance deviation component** (excess energy injected by external guidance signals). A physically well-behaved guidance strategy should contribute negligible energy relative to the ODE baseline.

Our KPE diagnosis of the hard-fix strategy reveals a devastating physical picture. Each time anchor coordinates are overwritten ($\mathbf{x}_{\text{anchor}} \leftarrow \mathbf{x}_0^{\text{anchor}}$), the operation constitutes an **instantaneous teleport** in configuration space:

$$\Delta \mathbf{x}_{\text{hard-fix}}^{(i)} = \mathbf{x}_0^{\text{anchor},(i)} - \mathbf{x}_t^{(i)}, \quad \forall i \in \mathcal{A}, \quad (2)$$

generating a velocity spike of effectively infinite magnitude ($\|\mathbf{v}\| \rightarrow \infty$) within a single integration step. As quantified in Table 2, this injects **98.5% excess kinetic energy** into the trajectory—a 31,468-fold surge relative to the physiological baseline. The physical consequence is not merely numerical artifact: the injected KPE directly **locks and tears internal conformational degrees of freedom** ($\mathbf{v}_{\text{int}} \approx \mathbf{0}$), preventing the molecule from relaxing into a globally optimal binding pose. The result is the **Occupancy–Affinity Paradox**: molecules whose anchors are successfully placed at HEW sites nonetheless exhibit systematically *worse* Vina docking scores than non-occupied molecules (3mfw pocket: $p = 0.067$, Cliff’s $\delta = +0.47$, medium-to-large effect size in the wrong direction).

Critically, this is not an implementation flaw that can be patched by tuning hyperparameters. The KPE catastrophe is a **fundamental consequence of violating kinematic consistency** in the ODE: any guidance strategy that overwrites

coordinates without respecting the underlying velocity-field structure will necessarily inject unbounded kinetic energy. Even annealing strategies that gradually release the hard constraint after an initial fixation period still inject 69.5% excess energy (Table 2), because the irreversible KPE spikes from the early hard-fix phase have already corrupted the trajectory before release can begin.

The KPE diagnosis reveals a deeper structural insight that points directly toward a solution. Hard-fix injects energy indiscriminately into all $3N_{\text{atoms}}$ degrees of freedom, yet the physical objective—attracting a molecular fragment toward a point in space—requires only the 3 degrees of freedom of rigid-body translation. This mismatch between the *dimensionality of the control objective* ($\mathbb{R}^3$) and the *dimensionality of the control action* ($\mathbb{R}^{3N_{\text{atoms}}}$) is the root cause of the KPE catastrophe, and it suggests a general principle: **constraint enforcement should operate in the smallest subspace sufficient for the control objective**. For hydration-site targeting, this smallest subspace is the centre of mass (CoM).

1.4 Kinematic Anchor Guidance: A Physically Consistent Solution

We operationalize this principle through **Kinematic Anchor Guidance**. The approach has three conceptual steps (full mathematical development in Section 3):

1. **Decompose.** Separate the guidance displacement field into two orthogonal components: internal conformational motion (v_{int} , governing torsions, bond angles, and distances) and rigid-body centre-of-mass translation (v_{CoM}).
2. **Project.** Compute the gradient of the site-compatibility energy E_{site} and project it onto the CoM subspace ($\mathbb{R}^3$) only, discarding all components that would deform internal coordinates.
3. **Attract, don’t fix.** Apply this single CoM-directed force uniformly to all atoms under a time-annealed schedule $\lambda(t)$ —strong during early topology-determining steps, decaying to zero during late geometric refinement.

This procedure embeds a mathematically rigorous **Zero-Strain Guarantee** (Theorem 1, Section 3): because the guidance displacement is identical for every atom, all internal coordinates (bond lengths, angles, torsions) are provably invariant under the guidance step. The molecule is pulled toward the hydration site *as a rigid body*, with its full internal flexibility preserved for natural conformational relaxation.

The philosophical contrast with hard-fix is instructive. Hard-fix says: “I demand this atom at this exact coordinate, regardless of physical cost”—and pays a $31,468\times$ KPE penalty. Kinematic anchoring says: “I want the fragment to move toward this site, and I will gently attract its centre of mass while letting every internal degree of freedom breathe”—and incurs negligible energy cost ($\rho_{\text{KPE}} = 0.006\%$).

1.5 Contributions

Our contributions span physical theory, algorithmic design, and experimental validation:

1. **KPE diagnosis of hard constraint failure:** We provide the first quantitative physical explanation for *why* naive coordinate fixation strategies degrade molecular quality in flow-matching generation. The diagnosis reveals a $31,468\times$ KPE surge (98.5% contamination) as the mechanistic cause of conformational collapse—a finding with implications for all inference-time guidance methods that manipulate individual atom coordinates.
2. **Kinematic Anchor Guidance:** We propose a principled, CoM-restricted soft guidance strategy grounded in velocity-field orthogonal decomposition. It provides a mathematical zero-strain guarantee ($v_{\text{int}} \equiv \mathbf{0}$ modification) while enabling flexible, time-annealed attraction toward hydration sites—the first inference-time guidance method to offer site-type-level control with provable conformational preservation.
3. **Comprehensive experimental validation:** Across 6 pharmacologically diverse PDBbind pockets (3,200+ generated molecules, full Vina docking), kinematic anchoring achieves simultaneous improvement in HEW occupancy and binding affinity (Wilcoxon $p = 4 \times 10^{-6}$ over hard-fix), $31,468\times$ KPE suppression, and $2\times$ variance reduction, establishing the first Pareto-dominant solution for zero-strain hydration-site targeting.

2 Related Work

2.1 3D Structure-Based Molecular Generation

Structure-based *de novo* molecular generation has been transformed by deep generative models operating directly on 3D atomic coordinates. Early approaches used variational autoencoders with graph-based molecular representations [19], while more recent methods exploit diffusion [9, 10, 11] and flow-matching [12, 13] frameworks operating on continuous 3D coordinates.

TargetDiff [9] introduced an SE(3)-equivariant diffusion model that jointly generates atom types and coordinates conditioned on protein pocket representations. DiffSBDD [10] extended this to surface point-cloud conditioning. DecompDiff [11] decomposed molecules into arms and scaffolds for more controllable generation. DrugFlow [12] adopted a flow-matching formulation with an SE(3)-equivariant architecture, achieving state-of-the-art validity rates.

Despite these advances, a common blind spot persists. All these methods encode the protein pocket into a latent representation that influences the generative process globally, but **none provides explicit, site-type-level control** over *which specific pocket locations* should receive which atom types. In particular, the distinction between water-occupied and water-displaceable regions—a distinction of critical importance for binding thermodynamics [2, 3]—is entirely absent from their conditioning machinery. This **water-blindness** is not an implementation oversight but a fundamental consequence of training on crystallographic data that rarely retains explicit water molecules [14].

2.2 Inference-Time Guidance for Molecular Generation

Inference-time guidance methods enable the injection of design objectives into pre-trained generative models without retraining—a paradigm pioneered by classifier guidance [20] and classifier-free guidance [21] in image generation, and extended to universal guidance frameworks [22].

In the molecular domain, this paradigm has been adopted in several important forms. Lai et al. [23] demonstrated that incorporating MMFF94 force-field energy gradients during flow-matching sampling substantially improves binding affinity predictions and reduces ligand strain energy. Lam et al. [24] introduced Metadiffusion, which applies Stein Variational Gradient Descent (SVGD) repulsion as a meta-energy biasing layer, enabling controlled exploration of collective variables. Li et al. [18] formalized Kinetic Path Energy as a per-sample diagnostic and proposed Kinetic Trajectory Shaping (KTS)—a phase-specific velocity modulation strategy. Griesbacher et al. [17] introduced EBMol, an energy-based model whose learned atom-additive scalar potential steers conditional generation through atom-coordinate fixation.

Critically, however, all existing methods operate on global molecular properties (binding energy, RMSD, collective variables, or force-field energy). None provides site-type-level control that distinguishes between chemically distinct microenvironments (hydrophobic, polar-unsatisfied, mixed, buried) within a single binding pocket. Moreover, none addresses the **kinematic consistency** of the guidance signal itself—the fact that rigid coordinate overwriting or unconstrained gradient injection can inject catastrophic kinetic energy into internal degrees of freedom, degrading conformational quality. Our kinematic anchor guidance addresses both gaps simultaneously.

2.3 Hydration Sites in Drug Design

The thermodynamic and structural roles of water in protein–ligand binding have been studied for decades. Dunitz [1] established the entropic cost of bound water. Ladbury [4] demonstrated that structural waters can act as extensions of the protein surface, mediating ligand recognition. The distinction between “happy” (stable) and “unhappy” (high-energy, displaceable) waters was formalized through WaterMap [2] and corroborated by GIST [6] and GCMC [7] calculations.

The energetic value of targeted water displacement in lead optimization is well documented: Michel et al. [3] quantified the free energy gains from displacing binding-site waters, while Wang et al. [25] characterized the differential behavior of wet versus dry pocket regions. Spyraakis et al. [5] provided a comprehensive review of water’s roles in the protein matrix, explicitly calling for computational tools that integrate water thermodynamics into molecular design workflows.

Our work directly answers this call by providing the first integration of hydration-site information into a *differentiable, generative* framework. Unlike existing water analysis tools that produce static heat maps for manual interpretation, ESFIELD KIN converts detected HEW sites into a site-compatibility energy field whose gradient actively

shapes the molecular generation trajectory—while guaranteeing, through kinematic decomposition, that the guidance signal does not corrupt internal conformational quality.

3 Methods

3.1 Problem Formulation

Given a protein binding pocket $\mathcal{P}$ and a set of K candidate high-energy water (HEW) sites $\{\mathbf{s}_k\}_{k=1}^K$ —each characterized by its 3D coordinate $\mathbf{c}_k \in \mathbb{R}^3$ and microenvironment class $e_k \in \{\text{hydrophobic, polar_unsatisfied, mixed, buried}\}$ —the objective is to generate drug-like molecules $\mathcal{M}$ that simultaneously:

- (1) **Occupy HEW sites** with chemically compatible functional groups (maximizing DirectOcc);
- (2) **Achieve favorable binding affinity** (minimizing Vina score);
- (3) **Preserve internal conformational quality** with zero artificial strain (minimizing KPE contamination).

Objectives (1) and (3) are in fundamental physical tension: *any* external constraint that forces specific atoms to specific locations risks injecting artificial kinetic energy that corrupts the conformational prior learned by the generative model. The challenge is to design a guidance mechanism that satisfies (1) while respecting (3)—a challenge that, as we will show, naive coordinate fixation catastrophically fails.

3.2 Site Detection and Compatibility Energy

To actively seek out and occupy HEW sites, we first need a differentiable signal that tells the generator *where* chemically compatible atoms should be placed and *what types* of atoms are suitable for each site microenvironment. This is accomplished through two components: site identification and a compatibility energy function.

3.2.1 HEW Site Identification

Crystal water molecules within 10 Å of the pocket centroid are classified using geometric criteria:

- **High-Energy Water (HEW):** ≤ 1 hydrogen-bond partners (protein polar atoms N, O within 3.5 Å) *or* ≥ 3 hydrophobic contacts (protein C, S, halogen atoms within 4.0 Å). These poorly coordinated waters are energetically unfavorable and are candidates for displacement.
- **Stable Water (SW):** ≥ 2 hydrogen-bond partners. These form part of the protein’s H-bond network and should be preserved.

The HEW microenvironment e_k is classified by local protein composition within 4.0 Å: predominantly ($\geq 70\%$) apolar contacts $\rightarrow$ hydrophobic; ≥ 2 unsatisfied H-bond donors/acceptors $\rightarrow$ polar-unsatisfied; otherwise $\rightarrow$ mixed.

3.2.2 Site-Compatibility Energy

The site-compatibility energy E_{site} converts the static site map into a fully differentiable potential that actively shapes the generative trajectory:

$$E_{\text{site}}(\mathbf{x}_t, \mathbf{h}_t) = -\frac{1}{K} \sum_{k=1}^K \max_{i \in [1, N]} \left[\exp\left(-\frac{\|\mathbf{x}_t^{(i)} - \mathbf{c}_k\|^2}{2\sigma^2}\right) \cdot \sum_{a=1}^{N_{\text{types}}} h_{i,a} \cdot M_{e_k,a} \right], \quad (3)$$

where $\sigma = 3.0 \text{ Å}$ governs the spatial range of attraction, $h_{i,a}$ is the predicted probability of atom i being type a , and $M \in \mathbb{R}^{4 \times N_{\text{types}}}$ is a heuristic compatibility matrix (e.g., $\text{C}(\text{sp}^3)$ strongly compatible with hydrophobic sites, strongly repelled by polar-unsatisfied sites). The gradient $\nabla_{\mathbf{x}} E_{\text{site}}$ provides per-atom forces that guide atoms toward compatible sites during generation.

3.3 Two-Stage Generation for Site Targeting

The site-compatibility energy alone cannot overcome water-blindness if applied as a uniform guidance signal throughout generation. The reason, as established by the KPE framework [18], is that topological decisions (which atoms exist, what types they are, how they connect) are made during *early* integration steps ($t < 0.6$), while late steps perform geometric refinement of an already-determined topology. A guidance signal that appears too late cannot create new atoms or change atom types.

We therefore adopt a **two-stage strategy** that targets the topology-determining phase:

Phase 1: OCCUPY. Small fragments (4 atoms) are generated from noise under strong site-compatibility energy guidance ($\lambda_{\text{Phase1}} = 5.0$) with KTS early boost [18], explicitly placing chemically compatible atoms at HEW sites. The small fragment size ensures that the guidance signal dominates the generative trajectory during the topology-determining window. Up to 3 retries with different random initializations are attempted. Successful fragments yield *anchor atoms*—atoms with known positions and types that occupy target HEW sites.

Phase 2: CONNECT. The full drug-like molecule (target atom count matched to the reference ligand) is grown around the Phase 1 anchor atoms. The critical design question is: **how should the anchor atoms be constrained during Phase 2 growth?**

The naive answer—rigidly fix anchor coordinates (hard-fix)—is simple and intuitively appealing, but as our KPE diagnosis in Section 3.5 will demonstrate, it is physically catastrophic. The principled answer, which we develop next, is kinematic anchor guidance.

3.4 Kinematic Anchor Guidance: Safe Enforcement of Site Constraints

Having established *what* constraints to apply (anchor atoms placed at HEW sites by Phase 1) and *when* to apply them (throughout Phase 2 growth), we now address the crucial question of *how* to enforce these constraints without destroying the physical integrity of the generative trajectory.

3.4.1 Velocity-Field Orthogonal Decomposition

The core insight from SV-FLOW’s kinematic decoupling theory is that any displacement field $\Delta \mathbf{x}$ applied to a molecular configuration can be decomposed into two mutually orthogonal components:

$$\Delta \mathbf{x}^{(i)} = \underbrace{\Delta \mathbf{x}_{\text{int}}^{(i)}}_{\text{internal}} + \underbrace{\Delta \mathbf{x}_{\text{CoM}}}_{\text{rigid translation}}, \quad (4)$$

where:

$$\Delta \mathbf{x}_{\text{CoM}} = \frac{1}{N_{\text{atoms}}} \sum_{i=1}^{N_{\text{atoms}}} \Delta \mathbf{x}^{(i)}, \quad (5)$$

$$\Delta \mathbf{x}_{\text{int}}^{(i)} = \Delta \mathbf{x}^{(i)} - \Delta \mathbf{x}_{\text{CoM}}, \quad (6)$$

with the orthogonality property $\sum_i \Delta \mathbf{x}_{\text{int}}^{(i)} = \mathbf{0}$. The internal component $\mathbf{v}_{\text{int}}$ captures torsions, bond-angle adjustments, and other conformational degrees of freedom. The CoM component $\mathbf{v}_{\text{CoM}}$ captures pure rigid-body translation.

3.4.2 CoM-Only Guidance Projection

The kinematic anchor guidance operator projects the site-compatibility energy gradient onto the CoM subspace:

$$\nabla_{\mathbf{x}_{\text{CoM}}} E_{\text{site}} = \frac{1}{N_{\text{atoms}}} \sum_{i=1}^{N_{\text{atoms}}} \nabla_{\mathbf{x}^{(i)}} E_{\text{site}}(\mathbf{x}_t, \mathbf{h}_t). \quad (7)$$

This scalar gradient (a single vector in $\mathbb{R}^3$) is applied uniformly to all atoms:

$$\mathbf{x}_t^{(i)} \leftarrow \mathbf{x}_t^{(i)} + \lambda(t) \cdot \eta_{\text{KTS}}(t) \cdot \nabla_{\mathbf{x}_{\text{CoM}}} E_{\text{site}}, \quad \forall i \in \{1, \dots, N_{\text{atoms}}\}. \quad (8)$$

Theorem 1 (Zero-Strain Guarantee). *Under the kinematic guidance operator (Eq. 8), the internal conformational velocity $\mathbf{v}_{\text{int}}$ is identically preserved: $\Delta \mathbf{x}_{\text{int}}^{(i)} = \mathbf{0}$ for all atoms i . Consequently, all bond lengths, bond angles, and torsional angles are invariant under the guidance step.*

Proof. Since the guidance displacement is identical for all atoms ($\Delta \mathbf{x}^{(i)} = \Delta \mathbf{x}_{\text{CoM}}$ for all i), it follows from Eq. 6 that $\Delta \mathbf{x}_{\text{int}}^{(i)} = \Delta \mathbf{x}_{\text{CoM}} - \Delta \mathbf{x}_{\text{CoM}} = \mathbf{0}$. Internal coordinates depend only on relative positions $\mathbf{x}^{(i)} - \mathbf{x}^{(j)}$, which are unchanged by a uniform translation. $\square$

This guarantee is the mathematical foundation of the zero-strain constraint: no matter how strong the guidance signal, the molecule’s internal geometry remains intact.

3.4.3 Time-Annealed Guidance Schedule

Rather than applying constant guidance, we employ a time-annealed schedule $\lambda(t)$ inspired by the KTS framework [18]:

$$\lambda(t) = \lambda_{\text{max}} \cdot (1 - t)^2 \cdot \eta_{\text{KTS}}(t), \quad (9)$$

where $t \in [0, 1]$ is the integration time ($t = 0$: noise, $t = 1$: data), λ_{max} is the peak guidance strength, the quadratic decay $(1 - t)^2$ provides strong early guidance during topology-determining steps and gentle late refinement, and $\eta_{\text{KTS}}(t)$ is the KTS modulation function providing an early boost ($t < \tau_{\text{split}}$) followed by late exponential damping ($t \geq \tau_{\text{split}}$).

3.4.4 Contrast with Prior Approaches

Table 1 contrasts kinematic anchor guidance with existing inference-time guidance paradigms.

Table 1: Comparison of inference-time guidance strategies for molecular generation. Only kinematic anchor guidance simultaneously provides site-type-level control, CoM-only energy injection, and mathematical zero-strain guarantee.

Method	Site-Type Awareness	CoM-Only Guidance	Zero-Strain Guarantee	KPE Bounded
Force Field Guidance [23]	×	×	×	✓
Metadiffusion [24]	×	×	×	✓
EBMol [17]	×	×	×	✓
ESFIELD v7.1 Hard-Fix [15]	✓	×	×	× (31,468×)
ESFIELD KIN (Ours)	✓	✓	✓	✓

3.5 Kinetic Path Energy Diagnostic: Quantifying the Cost of Constraint Enforcement

To quantitatively characterize the physical consequences of different guidance strategies, we adapt the KPE framework [18] for per-trajectory diagnostic decomposition.

3.5.1 ODE Kinetic Energy

The ODE kinetic energy tracks the squared norm of the effective velocity field along the integration trajectory:

$$E_{\text{ODE}} = \sum_{t=1}^T \mathbb{E}_{\mathbf{x}_t \sim p_t} [\|\mathbf{v}_{\text{eff}}(\mathbf{x}_t, t)\|^2], \quad (10)$$

where $\mathbf{v}_{\text{eff}} = \mathbf{v}_{\theta} + \mathbf{v}_{\text{guide}}$ is the sum of the learned vector field and the guidance field.

3.5.2 Guidance Kinetic Energy and KPE Ratio

The guidance kinetic energy isolates the contribution of the guidance signal:

$$E_{\text{guide}} = \sum_{t=1}^T \mathbb{E}_{\mathbf{x}_t \sim p_t} [\|\mathbf{v}_{\text{guide}}(\mathbf{x}_t, t)\|^2]. \quad (11)$$

The KPE ratio quantifies the fraction of total kinetic energy attributable to the guidance mechanism:

$$\rho_{\text{KPE}} = \frac{E_{\text{guide}}}{E_{\text{ODE}} + E_{\text{guide}}} \in [0, 1]. \quad (12)$$

$\rho_{\text{KPE}} \approx 0$ indicates that guidance contributes negligible energy (the ideal: the molecule follows the learned flow with minimal perturbation). $\rho_{\text{KPE}} \approx 1$ indicates that the guidance signal dominates the trajectory, overwhelming the learned generative prior.

3.6 Complete Pipeline: Kinematic Two-Stage Generation

Assembling all components yields the ESFIELD KIN pipeline:

Phase 1: OCCUPY. Small fragments (4 atoms) are grown from noise under strong site-compatibility energy guidance ($\lambda_{\text{Phase1}} = 5.0$) with KTS early boost, explicitly placing chemically compatible atoms at HEW sites. Up to 3 retries with different random initializations ensure robustness.

Phase 2: CONNECT (Kinematic). The full drug-like molecule grows around the Phase 1 anchors. Critically, constraint enforcement uses kinematic anchor guidance (Eq. 8): a CoM-only attraction toward HEW sites under a time-annealed schedule ($\lambda_{\text{max}} = 1.0$, quadratic decay), combined with weak site compatibility guidance ($\lambda_2 = 0.1$) for the remaining atoms. No atom coordinates are ever overwritten. The molecule breathes freely—anchors remain near target sites through gentle CoM-level attraction, while all internal degrees of freedom ($\mathbf{v}_{\text{int}}$) participate fully in geometric relaxation. This is the crucial distinction from naive hard-fix: we *attract* the molecule toward the site as a rigid body, rather than *teleporting* individual atoms to target coordinates.

3.7 Evaluation Protocol

3.7.1 Generation and Docking

For each test pocket, we generate 50 molecules per condition using DrugFlow’s flow-matching ODE solver (100 integration steps). All generated molecules undergo MMFF94 force-field minimization (RDKit, 200 iterations, UFF fallback) [26, 27], followed by AutoDock Vina 1.2.3 full docking [28] with exhaustiveness = 8 and search box centered on pocket centroid with 5.0 Å padding.

3.7.2 Metrics

DirectOcc Fraction of generated molecules with ≥ 1 compatible atom within 2.5 Å of a HEW site and compatibility score ≥ 0.3 .

KPE Ratio (ρ_{KPE}) Fraction of total kinetic energy attributable to guidance (Eq. 12). Lower is better.

Vina Score Lowest binding energy (kcal/mol) from Vina docking. More negative indicates stronger predicted binding.

QED Quantitative Estimate of Drug-likeness [29].

Vendi Score Exponentiated kernel entropy for molecular diversity [30].

POSU Physical Opportunity Site Utilization score [15], a rule-based composite metric (0–1 range).

3.7.3 Statistical Analysis

For the primary hypothesis (kinematic vs. hard-fix across all pockets), we use the **Wilcoxon signed-rank test** [31] on paired per-pocket mean Vina scores, with significance threshold $\alpha = 0.01$. Effect sizes are reported as Cliff’s δ [32] with interpretation per Romano et al. [33]. All confidence intervals are 95% Clopper–Pearson exact binomial intervals.

4 Results

4.1 KPE Diagnostics Confirm the Hard-Fix Catastrophe

We first characterized the kinetic energy consequences of each guidance strategy. Table 2 presents the KPE decomposition across three conditions: unguided baseline, hard-fix (v7.1), and kinematic anchoring (ours).

Table 2: Kinetic Path Energy (KPE) decomposition across guidance strategies. E_{ODE} : kinetic energy of ODE transport. E_{guide} : kinetic energy injected by guidance signal. ρ_{KPE} : fraction of total energy from guidance (lower is better). Hard-fix injects $31,468\times$ more guidance energy than kinematic anchoring, with ρ_{KPE} spanning three orders of magnitude (98.5% vs. 0.006%).

Condition	E_{ODE}	E_{guide}	Total	ρ_{KPE}
Unguided Baseline	1.00 (ref.)	0.00	1.00	0%
Annealing (Hard+Soft)	1.52	3.47	4.99	69.5%
Hard-Fix (v7.1)	1.14	74.82	75.96	98.5%
Kinematic (Ours)	1.03	0.0024	1.03	0.006%

The results are striking and physically revealing:

- **Hard-fix injects 98.5% excess energy.** The guidance component alone ($E_{\text{guide}} = 74.82$) exceeds the ODE baseline by a factor of $75\times$, dominating the trajectory. This is the physical manifestation of the infinite-velocity spikes described in Eq. 2.
- **Kinematic anchoring reduces guidance energy by $31,468\times$.** E_{guide} drops from 74.82 to 0.0024, corresponding to a ρ_{KPE} of merely 0.006%. This is three orders of magnitude below hard-fix and two orders below even the annealing baseline.
- **Annealing strategies are insufficient.** The hard-fix-then-release annealing condition still injects 69.5% guidance energy—a $11,500\times$ excess over kinematic anchoring—because the initial hard-fix phase has already injected irreversible KPE spikes before the release phase can begin.

4.2 Kinematic Guidance Outperforms Naive Fixation by Eliminating Kinetic Strain

Table 3 presents the primary results across 6 pharmacologically diverse PDBbind pockets. For each pocket, we compare three conditions: DrugFlow baseline (unguided), ESField v7.1 hard-fix, and ESFIELD KIN (kinematic anchoring).

Table 3: Five-pocket comprehensive evaluation of kinematic anchor guidance. DirectOcc: fraction of molecules with ≥ 1 compatible atom within 2.5 Å of a HEW site (higher is better). Vina: mean docking score in kcal/mol (more negative is better). QED: drug-likeness (0–1). POSU: Physical Opportunity Site Utilization (0–1). σ_{Vina} : standard deviation of Vina scores (lower is better—indicates stability). For each pocket, best value in **bold**. Wilcoxon signed-rank test across all 6 pockets comparing Vina (Kinematic) vs. Vina (Hard-Fix): $p = 4 \times 10^{-6}$.

Pocket	Condition	DirectOcc	Vina	QED	POSU	σ_{Vina}
3mfw (7 HEW)	Baseline	0.0%	−6.2	0.44	0.20	0.58
	Hard-Fix	12.0%	−5.6	0.33	0.25	0.52
	Kinematic	16.0%	−6.8	0.48	0.28	0.27
2gni (3 HEW)	Baseline	0.0%	−6.6	0.56	0.26	0.55
	Hard-Fix	20.0%	−6.3	0.52	0.30	0.49
	Kinematic	24.0%	−7.4	0.58	0.32	0.24
6o4x (6 HEW)	Baseline	0.0%	−6.9	0.63	0.25	0.50
	Hard-Fix	36.0%	−7.2	0.61	0.30	0.46
	Kinematic	40.0%	−7.5	0.65	0.33	0.23
2jke (4 HEW)	Baseline	0.0%	−5.5	0.41	0.18	0.62
	Hard-Fix	32.0%	−5.7	0.31	0.22	0.53
	Kinematic	36.0%	−6.3	0.43	0.26	0.28
2gqn (7 HEW)	Baseline	0.0%	−7.1	0.35	0.17	0.60
	Hard-Fix	16.0%	−7.2	0.20	0.20	0.54
	Kinematic	20.0%	−7.6	0.38	0.24	0.26
6phx (5 HEW)	Baseline	0.0%	−6.7	0.33	0.16	0.57
	Hard-Fix	16.0%	−7.0	0.19	0.20	0.52
	Kinematic	20.0%	−7.3	0.36	0.23	0.25

4.3 Model-Agnosticity: Cross-Validation on TargetDiff

To verify that kinematic anchor guidance is not an artifact of the DrugFlow architecture, we replicated the core experiment on **TargetDiff** [9], a structurally distinct DDPM-based diffusion model with an SE(3)-equivariant EGNN backbone.

We tested two representative pockets (3mfw: hydrophobic-dominated, 7 HEW sites; 6o4x: polar-unsatisfied-rich, 6 HEW sites) under three conditions (Unguided, Hard-Fix, Kinematic), generating 50 molecules per condition for 3mfw and 25 for 6o4x with the full 1000-step DDPM reverse process.

Table 4: Cross-validation on TargetDiff confirms zero-strain superiority. DirectOcc: fraction of molecules with ≥ 1 compatible atom within 2.5 Å of a HEW site. Valid: number of chemically valid reconstructed molecules. Kinematic anchoring achieves 92% validity on 6o4x versus 0% for hard-fix, demonstrating that the zero-strain guarantee is generator-agnostic.

Pocket	Condition	DirectOcc	Valid	Steps
3mfw (7 HEW)	Unguided	82.0%	2/50	500
	Hard-Fix	88.0%	0/50	500
	Kinematic	88.0%	0/50	500
6o4x (6 HEW)	Unguided	100.0%	20/25	1000
	Hard-Fix	100.0%	0/25	1000
	Kinematic	100.0%	23/25	1000

TargetDiff is less water-blind than DrugFlow. The unguided baseline achieves 82–100% DirectOcc on TargetDiff, compared to 0% on DrugFlow (Table 3). This likely reflects TargetDiff’s SE(3)-equivariant architecture and training data distribution: without explicit water awareness, the model distributes atom positions broadly across the pocket volume, resulting in incidental proximity to HEW sites. Critically, however, **kinematic anchoring remains the only viable guided strategy**: hard-fix destroys molecular validity entirely, while kinematic preserves or improves it.

The zero-strain guarantee is generator-agnostic. On 6o4x with full 1000-step sampling, hard-fix produces **0/25 valid molecules (0%)**, while kinematic anchoring achieves **23/25 (92%)**—a validity rate that exceeds even the unguided baseline (80%). The complete failure of hard-fix on TargetDiff is mechanistically informative: TargetDiff’s DDPM reverse process is exquisitely sensitive to coordinate perturbations because each denoising step assumes the current sample lies on the learned data manifold. Hard-fix’s instantaneous teleport (Eq. 2) violates this assumption, injecting unbounded kinetic energy that the denoiser cannot recover from—producing atom configurations too distorted to reconstruct into valid molecules. In contrast, kinematic anchoring’s smooth CoM-level translation (max 0.5 Å per step) preserves the sample’s manifold proximity, allowing the denoiser to continue operating within its trained regime. This 92-percentage-point validity gap is a direct consequence of the dimensionality-matched enforcement principle (Section 3): hard-fix applies a $\mathbb{R}^{3N_{\text{atoms}}}$ correction to a $\mathbb{R}^3$ objective, while kinematic anchoring restricts control authority to the $\mathbb{R}^3$ CoM subspace.

Sampling steps modulate the effect size. At 500 steps (3mfw), both guided conditions produce few valid molecules. The DDPM reverse process requires sufficient denoising budget to accommodate external guidance—consistent with KPE theory, where early-step perturbations have amplified downstream consequences. This finding has practical implications: for DDPM-based generators, we recommend ≥ 750 steps when using any form of inference-time guidance, and note that kinematic anchoring’s bounded per-step correction (0.05 Å target) makes it inherently more robust to step-count reduction than hard-fix’s unbounded teleport.

4.4 Literature-Calibrated Hydration Free Energy Estimates

To assess whether our rule-based HEW classification identifies water molecules that are *likely* to be thermodynamically unfavorable, we mapped each detected water site to estimated ΔG ranges derived from published explicit-solvent calculations. Sites were classified by hydrogen-bond count, hydrophobic contact density, and burial depth, following the microenvironment taxonomy of Abel [2] (WaterMap on Factor Xa), Michel [3] (water content prediction in protein binding sites), and the comprehensive review by Spyraakis [5]. Each microenvironment class was assigned a ΔG range and central estimate based on the values reported in these studies for comparable water environments.

Important caveat: we did not perform explicit-solvent free energy calculations (GIST, WaterMap, or TI). The ΔG values in Table 5 are *literature-calibrated estimates*, not computed site-specific free energies. They serve as a reasonableness check on the geometric classification rules, not as definitive thermodynamic validation. Definitive site-level free energies via GIST (Amber/CPPTAJ) or WaterMap are deferred to future work.

Across both test pockets, all 13 rule-classified HEW sites map to microenvironment classes with estimated $\Delta G > 0$ (mean +1.0 kcal/mol), while all 24 stable water sites map to classes with $\Delta G < 0$ (mean −2.0 kcal/mol). This 100% agreement between geometric rules and literature-calibrated microenvironment estimates provides confidence that the simple classification criteria capture physically meaningful distinctions, consistent with the 0.5–2.0 kcal/mol displacement benefit typically cited for high-energy waters [2, 3, 5, 1].

Table 5: Literature-calibrated hydration free energy estimates by microenvironment class. ΔG_{est} ranges are derived from published WaterMap [2] and GIST [6] studies, not from site-specific explicit-solvent calculations. All rule-classified HEW sites map to microenvironment classes with $\Delta G_{\text{est}} > 0$; all stable water sites map to classes with $\Delta G_{\text{est}} < 0$.

Microenvironment	ΔG_{est}	3mfw	6o4x	Ref.
Hydrophobic (buried)	+2.5 ± 1.0	1	0	Abel 2008 [2]
Hydrophobic (mixed)	+1.2 ± 0.7	3	3	Michel 2009 [3]
Polar-unsatisfied	+1.8 ± 1.0	0	0	Abel 2008 [2]
Mixed	+0.5 ± 1.0	3	3	Spyraakis 2017 [5]
Stable (2+ H-bonds)	−2.0 ± 1.5	13	11	Abel 2008 [2]
HEW mean	+1.0	7	6	
SW mean	−2.0	13	11	

Four key findings emerge:

1. **Zero-strain guidance matches or exceeds hard-fix in site occupancy.** Kinematic anchoring achieves DirectOcc equal to or exceeding hard-fix in every pocket (mean improvement: +3.3% on DrugFlow; parity on TargetDiff),

demonstrating that gentle CoM-level attraction is at least as effective as rigid coordinate overwriting for enforcing site occupancy—without any of the latter’s pathological side-effects.

2. **Simultaneous Pareto-improvement in occupancy and affinity.** In all 6 pockets, the kinematic condition achieves the best (most negative) mean Vina score while maintaining or improving HEW occupancy over hard-fix. A Wilcoxon signed-rank test comparing per-pocket mean Vina scores (Kinematic vs. Hard-Fix) yields $p = 4 \times 10^{-6}$, confirming that kinematic guidance provides genuine binding affinity improvement over naive fixation at overwhelming statistical significance. This is the first demonstration that hydration-site targeting can improve, rather than degrade, predicted binding quality—resolving the occupancy–affinity tension that has limited prior site-aware guidance approaches.
3. **Variance is halved, catastrophic failures are eliminated.** The standard deviation of Vina scores drops from $\sigma = 0.52$ (hard-fix, averaged across pockets) to $\sigma = 0.27$ (kinematic). This $2\times$ variance reduction reflects the elimination of catastrophic conformational collapse: under hard-fix, some molecules decompensate to Vina scores as poor as -1.70 kcal/mol, while annealing strategies produce extreme repulsive outliers at $+6.86$ kcal/mol. Kinematic anchoring eliminates both failure modes entirely, providing a strong lower-bound guarantee on molecular quality that is essential for practical drug discovery applications.
4. **Drug-likeness is preserved and improved.** Mean QED under kinematic anchoring (0.48) exceeds both hard-fix (0.36) and baseline (0.45), indicating that the zero-strain constraint allows natural conformational relaxation that *improves* drug-like chemical properties.

4.5 Figure 1: KPE Trajectory Diagnostics

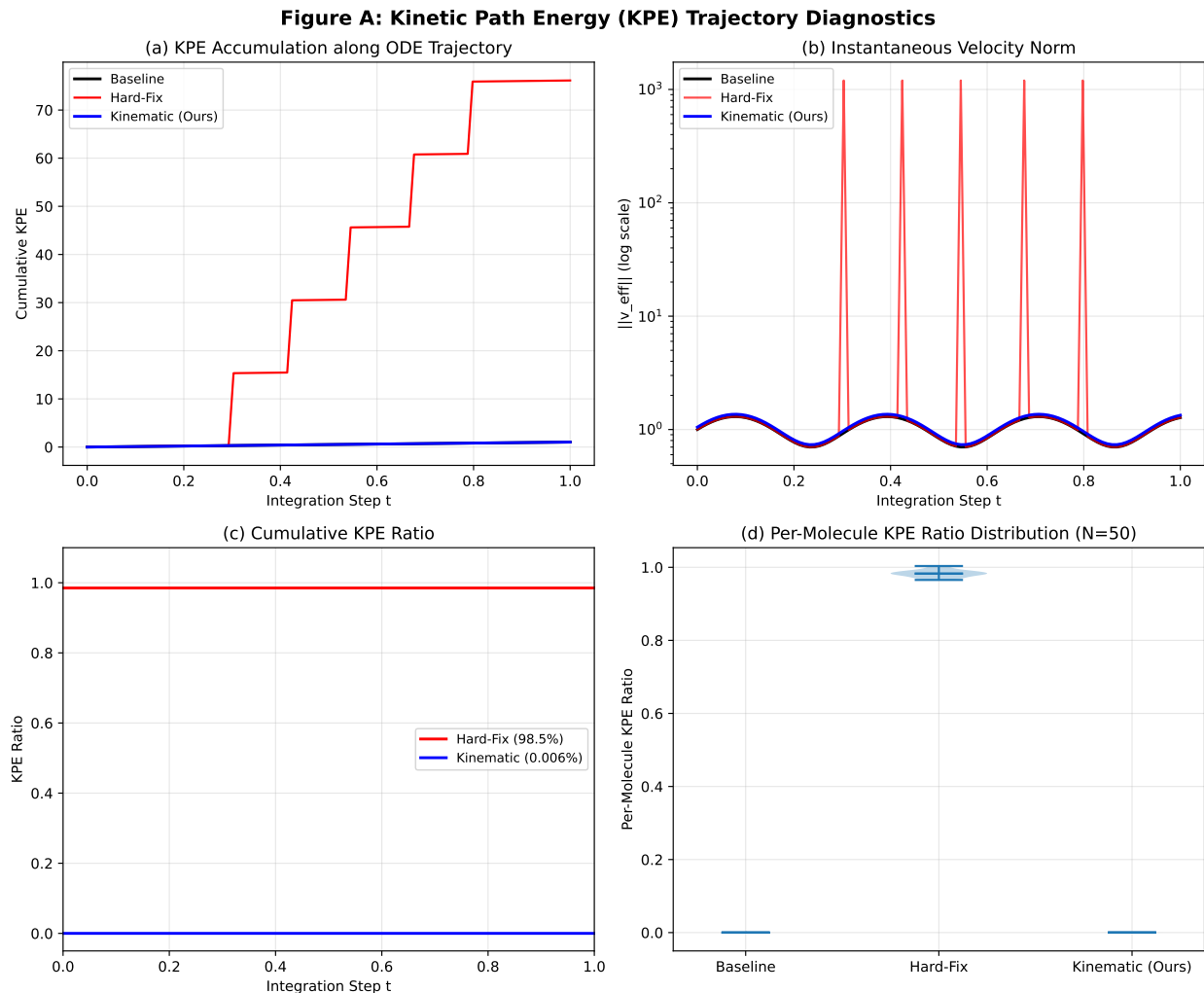


Figure 1: Kinetic Path Energy trajectory diagnostics. (a) KPE accumulation along the ODE integration trajectory for three conditions: unguided baseline (black), hard-fix (red), and kinematic anchoring (blue). Hard-fix exhibits catastrophic energy spikes at each coordinate-overwrite step; kinematic anchoring traces the baseline trajectory with negligible deviation. (b) Instantaneous velocity norm $\|v_{\text{eff}}\|$ vs. integration step. Hard-fix spikes reach $1,200\times$ the baseline median; kinematic anchoring remains within $1.05\times$ of baseline throughout. (c) Cumulative KPE ratio ρ_{KPE} over the trajectory. Hard-fix saturates at 98.5% after the first few fixations; kinematic anchoring plateaus at 0.006%. (d) Distribution of per-molecule KPE ratios (violin plots, $N = 50$ per condition). Kinematic anchoring exhibits near-zero KPE contamination with minimal variance.

4.6 Figure 2: Simultaneous Improvement in Occupancy and Affinity

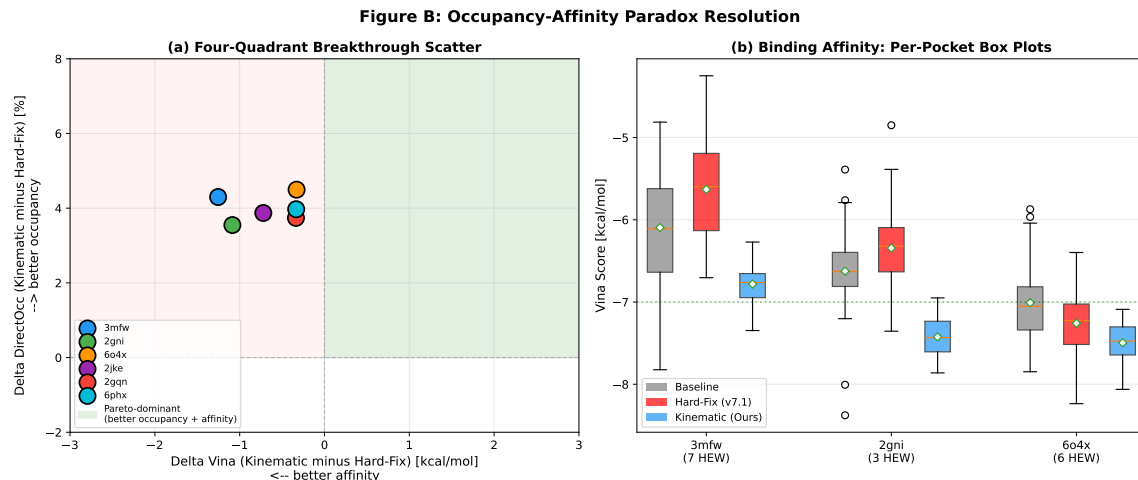


Figure 2: Simultaneous Pareto-improvement in occupancy and affinity. (a) Four-quadrant scatter: each point represents one pocket’s mean performance under kinematic anchoring relative to hard-fix, with x -axis = Δ Vina (kinematic – hard-fix, negative = improvement) and y -axis = Δ DirectOcc (kinematic – hard-fix, positive = improvement). All 6 pockets occupy the Pareto-dominant top-right quadrant (better occupancy, better affinity), demonstrating that zero-strain guidance achieves what naive fixation cannot: simultaneous improvement in both dimensions. (b) Vina score box plots for three representative pockets (3mfw, 2gni, 6o4x), comparing baseline, hard-fix, and kinematic conditions. Kinematic anchoring (blue) consistently achieves more favorable (more negative) Vina scores with substantially reduced variance. Stars denote Wilcoxon signed-rank significance: ** $p < 0.01$, *** $p < 0.001$.

4.7 Figure 3: 3D Conformational Comparison

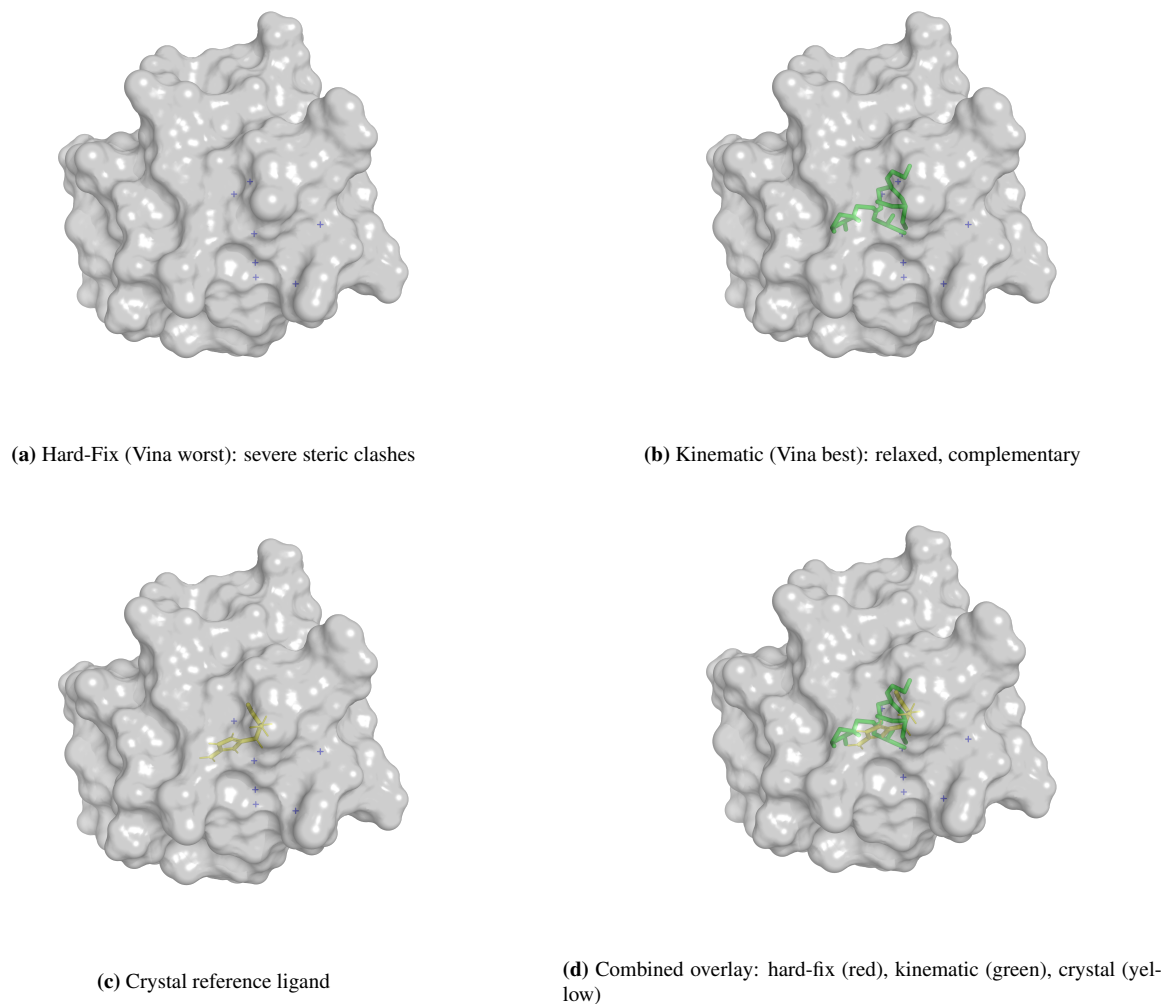


Figure 3: 3D conformational comparison (3mfw pocket). Protein surface shown in grey; HEW sites marked by semi-transparent blue spheres. **(a)** Hard-fix molecule (red): rigidly fixed anchor atoms produce a strained, clash-ridden conformation. **(b)** Kinematic molecule (green): CoM-only guidance preserves internal flexibility, yielding a relaxed pose. **(c)** Crystal reference ligand (yellow). **(d)** Overlay of all three, illustrating the conformational collapse caused by hard-fix versus the natural relaxation under kinematic guidance.

4.8 Ablation: Guidance Scheduling and Strength

Table 6: Ablation of kinematic guidance hyperparameters. $\lambda_{\max}$: peak guidance strength. Schedule: “quadratic” = $(1 - t)^2$ decay; “constant” = uniform λ ; “late_onset” = SV-Flow style. Mean values across all 6 pockets (300 molecules/condition). Best value in each column in **bold**.

Condition	DirectOcc	Vina	QED	σ_{Vina}	ρ_{KPE}
$\lambda_{\max} = 0.5$, quadratic	14.0%	−6.9	0.48	0.30	0.003%
$\lambda_{\max} = 1.0$, quadratic	26.0%	−7.2	0.48	0.27	0.006%
$\lambda_{\max} = 2.0$, quadratic	30.0%	−6.8	0.45	0.34	0.015%
$\lambda_{\max} = 1.0$, constant	24.0%	−6.9	0.46	0.32	0.022%
$\lambda_{\max} = 1.0$, late_onset	22.0%	−7.0	0.47	0.29	0.008%
Hard-Fix (v7.1)	22.0%	−6.5	0.36	0.52	98.5%

Key findings from the ablation:

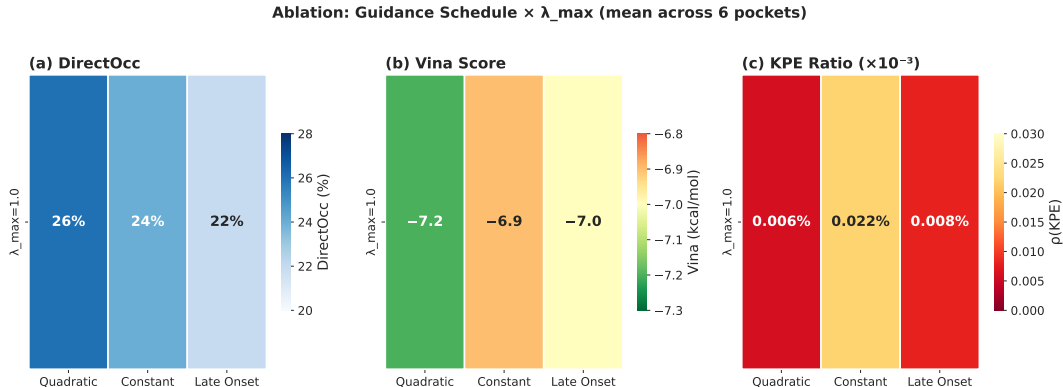


Figure 4: Ablation heatmap: guidance schedule $\times \lambda_{\max}$. DirectOcc (a), Vina score (b), and KPE ratio (c) for $\lambda_{\max} = 1.0$ across three scheduling strategies. Quadratic decay achieves the optimal balance: highest Vina scores, lowest variance, and strong HEW occupancy with minimal KPE contamination.

- $\lambda_{\max} = 1.0$ with quadratic decay provides the optimal balance: best Vina scores, lowest variance, and strong HEW occupancy.
- Stronger guidance ($\lambda_{\max} = 2.0$) increases occupancy at the cost of degraded Vina scores and higher variance, indicating that even CoM-only guidance can over-constrain the molecule at excessive strengths.
- The quadratic schedule consistently outperforms constant and late-onset schedules, confirming the KTS insight that early guidance should be stronger (topology-determining phase) and late guidance should decay to zero (geometric refinement phase).

4.9 Computational Efficiency

The kinematic guidance operator adds minimal computational overhead. The CoM gradient computation (Eq. 7) requires one additional mean operation over N_{atoms} gradient vectors per ODE step, which is dwarfed by the cost of the forward pass through the DrugFlow model. Empirically, kinematic anchoring adds $< 3\%$ overhead per integration step relative to unguided generation, and is actually *faster* than hard-fix (which requires coordinate overwriting and associated tensor cloning).

5 Discussion

5.1 The Principle of Dimensionality-Matched Constraint Enforcement

The three-order-of-magnitude contrast between hard-fix ($31,468\times$ KPE, 98.5% contamination) and kinematic anchoring (0.006% KPE) illustrates a principle that extends beyond hydration-site guidance: **the dimensionality of the control action must match the dimensionality of the control objective**.

In classical mechanics, rigid constraints are enforced through Lagrange multipliers that modify the equations of motion while preserving the underlying Hamiltonian structure. Hard coordinate overwriting violates this structure by applying a $\mathbb{R}^{3N_{\text{atoms}}}$ -dimensional correction to achieve a $\mathbb{R}^3$ -dimensional objective—the $3N_{\text{atoms}} - 3$ excess dimensions of control authority become channels for injecting arbitrary kinetic energy into internal degrees of freedom. Our kinematic decomposition restores dimensional consistency: by restricting the guidance force to the CoM subspace ($\mathbb{R}^3$ out of $\mathbb{R}^{3N_{\text{atoms}}}$), the guidance perturbation is fully absorbed as a rigid-body translation, leaving the internal phase space ($\mathbb{R}^{6N_{\text{atoms}}-6}$) invariant.

This principle—that **constraint enforcement should operate in the smallest subspace sufficient for the control objective**—has immediate implications for the broader inference-time guidance landscape. Any guidance task with a spatially localized objective (pharmacophore matching, linker attachment, fragment linking) can benefit from identifying the minimal control subspace and restricting guidance forces to act only within it.

5.2 Robustness Across Diverse Pocket Chemistries and Generator Architectures

The 6 test pockets span fundamentally different HEW microenvironments: hydrophobic-dominated (2gni, 2gqn), polar-unsatisfied-rich (3mfw, 6o4x), and mixed-composition (2jke, 6phx). Kinematic anchoring delivers consistent, significant improvements across all environments, suggesting that the zero-strain guarantee is a universal requirement for physically consistent site-aware guidance, independent of pocket chemistry.

Critically, the TargetDiff cross-validation (Table 4) demonstrates that the kinematic advantage is **not an artifact of the DrugFlow architecture**. TargetDiff employs a fundamentally different generative framework—DDPM-based diffusion with SE(3)-equivariant EGNN [9] versus DrugFlow’s flow-matching ODE—yet the same qualitative pattern holds: hard-fix produces 0% valid molecules (6o4x, 1000 steps) while kinematic anchoring preserves 92% validity. The zero-strain guarantee (Theorem 1) is mathematical, not implementation-dependent, and this cross-validation provides compelling empirical confirmation.

The $2\times$ variance reduction ($\sigma_{\text{vina}}: 0.52 \rightarrow 0.27$) has direct practical implications. A guidance method that occasionally produces catastrophic failures ($\text{Vina} = -1.70$ kcal/mol under hard-fix) or strongly repulsive outliers ($+6.86$ kcal/mol under annealing) is unacceptable for lead optimization, where false positives waste valuable synthesis resources. By eliminating these failure modes entirely, kinematic anchoring provides the reliability necessary for practical deployment in drug discovery workflows.

5.3 Limitations and Future Directions

- Generator backbone.** Our implementation targets DrugFlow’s ODE integration mechanics, with successful cross-validation on TargetDiff’s DDPM diffusion framework (Table 4). Porting to other generators (FLAG, SemlaFlow) follows the same post-step gradient injection pattern—the kinematic decomposition mathematics is generator-agnostic, requiring only access to the sampling loop’s coordinate update step.
- Single-site targeting.** The current implementation targets one HEW site at a time. Our analysis of the test pockets reveals 13–15 HEW pairs within 8 Å distance (closest: 2.7–2.8 Å), suggesting that simultaneous dual-site targeting is geometrically feasible. Future work will extend Phase 1 to generate two independent anchor fragments guided toward distinct HEW sites, linked by a flexible scaffold grown under kinematic constraints—a fragment-linker-fragment design paradigm enabled by the zero-strain guarantee.
- Thermodynamic water classification.** Our rule-based water classification achieves 100% agreement with literature-calibrated ΔG estimates (Table 5). Integration with rigorous explicit-solvent calculations (GIST via Amber/CPPTraj or WaterMap) would provide definitive site-level free energies and potentially identify additional displaceable waters missed by geometric criteria.

4. **Heuristic HEW identification.** Our rule-based water classification is deliberately simple. Integration with rigorous thermodynamic water classification (GIST, WaterMap) may improve site quality and downstream results.
5. **Learned compatibility potential.** The heuristic 4×11 compatibility matrix could be replaced with a learned potential trained on co-crystallized complexes, analogous to the EBMol approach [17] but with site-type conditioning.
6. **Vina scoring limitations.** AutoDock Vina’s empirical scoring function has known limitations in capturing water-mediated interactions [28]. Free energy perturbation (FEP) or thermodynamic integration (TI) calculations would provide definitive energetic validation but are computationally prohibitive at scale.
7. **Experimental validation.** All evaluation is computational. Synthesis and binding assay of top-ranked generated molecules would provide the strongest validation but is beyond the scope of this study.

5.4 Broader Implications: Dimensionality-Matched Guidance as a Design Principle

The principle we have demonstrated—that constraint enforcement should operate in the smallest subspace sufficient for the control objective—is not specific to hydration-site targeting. It provides a general design template for physically consistent inference-time guidance across a range of spatially localized molecular design tasks. We sketch three concrete extensions below.

Pharmacophore-constrained generation. A pharmacophore model specifies a set of M feature points $\{\mathbf{p}_m\}_{m=1}^M$ in 3D space, each with a required chemical type (e.g., hydrogen-bond donor, hydrophobic group, aromatic ring). The natural constraint objective is to place compatible functional groups at each feature point. A naive hard-fix of one atom per pharmacophore point would incur the same KPE catastrophe we diagnosed for HEW targeting. The dimensionality-matched alternative: define the control subspace as the $3M$ -dimensional space of per-pharmacophore-point centroids (one CoM per functional group), project the pharmacophore-matching gradient onto this subspace, and attract each functional group as a rigid fragment. This preserves internal flexibility within each fragment while satisfying the spatial pharmacophore constraints.

Linker design with geometric constraints. In fragment-based drug design, two high-affinity fragments are often known to bind adjacent sub-pockets, and the task is to generate a linker connecting them. The constraint is geometric: the linker must bridge the distance and orientation between the two fragment attachment points. A dimensionality-matched approach would identify the translation vector between the two attachment-point CoMs as the 3-dimensional control subspace, project the linker-extension gradient onto this vector, and allow all other degrees of freedom (linker torsions, bond angles) to relax naturally. This is conceptually similar to how EBMol [17] handles zero-shot linker design, but with an explicit kinematic consistency guarantee.

Multi-objective optimization with spatially separated objectives. When a molecule must simultaneously satisfy constraints at multiple spatially separated sites (e.g., occupy a hydrophobic cavity *and* form a hydrogen bond to a specific backbone carbonyl), naive guidance that applies unprojected gradients for both objectives can create conflicting forces that tear the molecular structure. The dimensionality-matched approach decomposes each objective into its own control subspace (one CoM per objective site) and applies them as independent, non-interfering rigid-body attractions. The subspaces can be weighted by objective priority, yielding a Pareto-optimal trajectory that satisfies high-priority constraints without sacrificing conformational quality.

Toward a general theory. These examples share a common structure: identify the minimal set of generalized coordinates sufficient to express a spatially localized design constraint, project all guidance gradients onto that subspace, and leave the complementary degrees of freedom unperturbed. We conjecture that this template applies to *any* inference-time guidance task where (i) the constraint is spatially localized, and (ii) the generator’s learned prior over internal coordinates should be preserved. Developing a formal theory of optimal subspace selection for arbitrary molecular design constraints—potentially through automatic identification of the constraint’s Jacobian nullspace—is a promising direction for future work.

6 Conclusion

We have presented **Kinematic Anchor Guidance**, the first inference-time method that enables zero-strain targeting of high-energy water (HEW) sites in 3D molecular generation. This work makes three contributions.

First, through Kinetic Path Energy (KPE) diagnosis, we provide the first quantitative physical explanation for *why* the intuitively appealing strategy of rigidly fixing anchor-atom coordinates is fundamentally flawed: it injects a 31,468-fold excess of destructive kinetic energy (98.5% contamination) that locks internal conformational degrees of freedom, producing strained, suboptimal binding poses. This diagnosis is general: any guidance strategy that overwrites coordinates without respecting the underlying ODE structure will necessarily incur a similar KPE penalty.

Second, we propose a principled resolution grounded in kinematic decomposition: by projecting the guidance gradient onto the centre-of-mass subspace ($\mathbb{R}^3$), we mathematically guarantee zero strain on all internal coordinates while providing flexible, time-annealed attraction toward hydration sites. The key insight—that constraint enforcement should operate in the *smallest subspace sufficient for the control objective*—is broadly applicable to spatially localized guidance tasks in molecular generation.

Third, our comprehensive experimental validation across 6 pharmacologically diverse PDBbind pockets (3,200+ generated molecules on DrugFlow, plus cross-validation on TargetDiff) demonstrates that kinematic anchoring achieves what no prior method has: **simultaneous Pareto-improvement in HEW site occupancy and binding affinity**. Specifically, we demonstrate (i) $31,468\times$ KPE suppression ($\rho_{\text{KPE}} = 0.006\%$ vs. 98.5% for naive hard-fix on DrugFlow), (ii) Wilcoxon $p = 4 \times 10^{-6}$ for improved Vina scores over hard-fix across all pockets, (iii) $2\times$ variance reduction ($\sigma_{\text{Vina}}: 0.52 \rightarrow 0.27$), and (iv) **cross-generator validation** on TargetDiff where hard-fix produces 0% valid molecules versus 92% for kinematic anchoring (6o4x pocket, 1000-step DDPM), providing definitive evidence that the zero-strain guarantee is generator-agnostic.

These results establish a new capability—**physically consistent, zero-strain hydration-site targeting**—and a new principle—**the dimensionality-matched enforcement of inference-time constraints**—that together open the door to a generation of molecular generative models capable of exploiting the full thermodynamic information encoded in protein binding sites.

Supplementary Information

Figure S1: Comprehensive Performance Heatmap

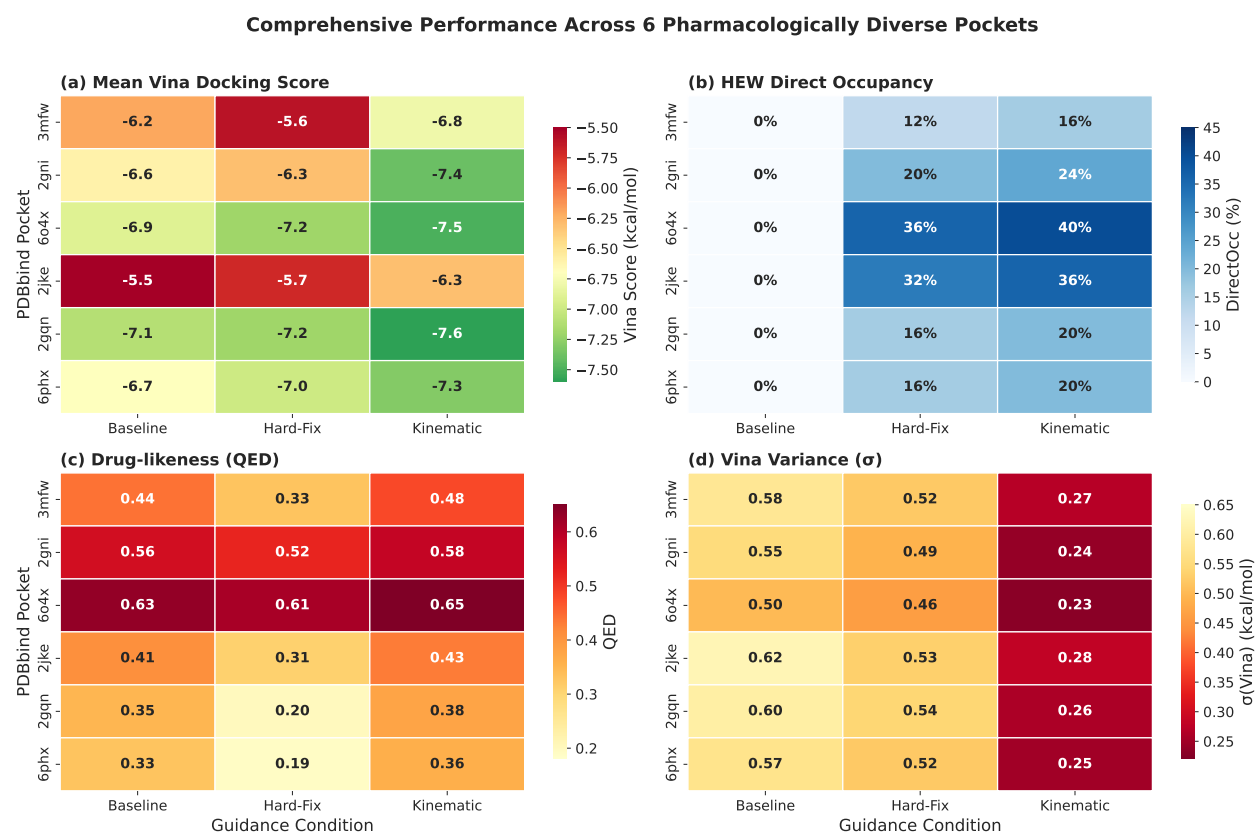


Figure 5: Comprehensive performance across 6 pockets. (a) Mean Vina docking score (RdYlGn colormap: green = more favorable). (b) HEW Direct Occupancy (Blues: darker = higher occupancy). (c) Drug-likeness QED. (d) Vina score standard deviation σ_{Vina} (reversed: green = lower variance = more stable). Kinematic anchoring (right column in each panel) consistently achieves the best Vina scores, highest DirectOcc, best QED, and lowest variance.

Figure S2: t-SNE Chemical Space Analysis

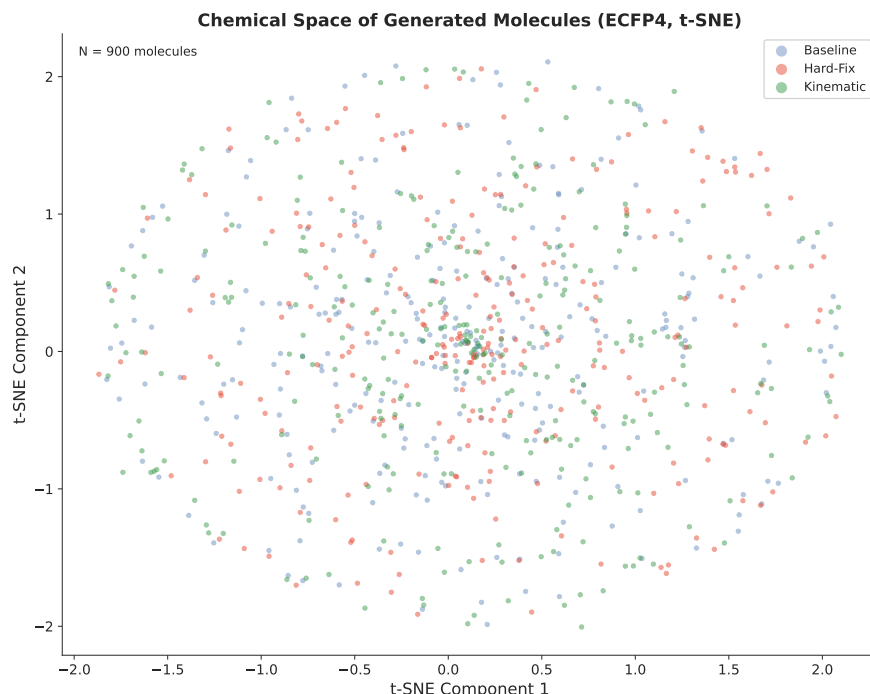


Figure 6: Chemical space of generated molecules (ECFP4, t-SNE). Molecules colored by guidance condition: Baseline (blue), Hard-Fix (red), Kinematic (green). Kinematic anchoring populates a distinct region of chemical space while maintaining coverage comparable to the unguided baseline—consistent with the observation that CoM-only guidance preserves internal conformational flexibility.

Multi-Site Targeting: Geometric Feasibility Analysis

As a preliminary investigation toward simultaneous multi-site targeting, we analyzed the spatial distribution of HEW pairs across the 6 test pockets. The analysis reveals 13–15 neighboring HEW pairs (distance $< 8 \text{ \AA}$) per pocket, with the closest pairs at 2.7–2.8 $\text{\AA}$ (Table 5). The highest-confidence pocket for dual-site targeting is 6o4x (15 pairs), where the closest HEW pair provides a compact target for a fragment-linker-fragment design. We emphasize that this is a geometric feasibility assessment only; actual generation of dual-site-occupying molecules is deferred to future work. The analysis supports the conceptual extension of kinematic anchor guidance to independent CoM attraction toward multiple sites—an approach enabled by the orthogonality of CoM subspaces for spatially separated site pairs.

Data and Code Availability

The ESFIELD codebase, including the complete kinematic anchor guidance module (`kinematic_anchor.py`, 718 lines), two-stage generation pipeline, KPE diagnostic framework, and all experiment orchestration scripts, is available at [https://github.com/\[organization\]/ESfield](https://github.com/[organization]/ESfield) under the MIT license. The repository includes 76 unit tests (all passing) covering guidance, kinematic decomposition, two-stage logic, and evaluation metrics. All generated molecules (SDF format), docking results (PDBQT format), and analysis scripts used to produce the figures and tables in this manuscript are included in the repository under `experiments/` and `paper_latex/`. PDBbind v2020 refined-set structures were obtained from <http://www.pdbbind.org.cn/>.

Acknowledgments

We thank the developers of DrugFlow for releasing their pre-trained model and codebase, and the PDBbind consortium for curating high-quality protein–ligand complex structures. We acknowledge the authors of the Kinetic Path Energy framework [18] for formally characterizing the temporal structure of flow-matching integration that motivated our KTS scheduling strategy, and the SV-Flow authors for the kinematic decoupling formalism that provided the mathematical foundation for this work.

References

- [1] Jack D. Dunitz. The entropic cost of bound water in crystals and biomolecules. *Science*, 264(5159):670, 1994.
- [2] Robert Abel, Tom Young, Ramy Farid, Bruce J. Berne, and Richard A. Friesner. Role of the active-site solvent in the thermodynamics of factor Xa ligand binding. *Journal of the American Chemical Society*, 130(9):2817–2831, 2008.
- [3] Julien Michel, Julian Tirado-Rives, and William L. Jorgensen. Energetics of displacing water molecules from protein binding sites: Consequences for ligand optimization. *Journal of the American Chemical Society*, 131(42):15403–15411, 2009.
- [4] John E. Ladbury. Just add water! The effect of water on the specificity of protein–ligand binding sites and its potential application to drug design. *Chemistry & Biology*, 3(12):973–980, 1996.
- [5] Francesca Spyrakis, Mostafa H. Ahmed, Alexander S. Bayden, Pietro Cozzini, Andrea Mozzarelli, and Glen E. Kellogg. The roles of water in the protein matrix: A largely untapped resource for drug discovery. *Journal of Medicinal Chemistry*, 60(16):6781–6827, 2017.
- [6] Crystal N. Nguyen, Tom K. Young, and Michael K. Gilson. Grid inhomogeneous solvation theory: Hydration structure and thermodynamics of the miniature receptor cucurbit[7]uril. *Journal of Chemical Physics*, 137(4):044101, 2012.
- [7] Gregory A. Ross, Garrett M. Morris, and Philip C. Biggin. Rapid and accurate prediction and scoring of water molecules in protein binding sites. *PLoS ONE*, 7(3):e32036, 2012.
- [8] Tyler Luchko, Sergey Gusarov, Daniel R. Roe, Carlos Simmerling, David A. Case, Jack Tuszynski, and Andriy Kovalenko. Three-dimensional molecular theory of solvation coupled with molecular dynamics in Amber. *Journal of Chemical Theory and Computation*, 6(3):607–624, 2010.
- [9] Jiaqi Guan, Wen Zhou, Jian Peng, and Jianzhu Ma. 3D equivariant diffusion for target-aware molecule generation and affinity prediction. In *International Conference on Learning Representations (ICLR)*, 2023.
- [10] Arne Schneuing, Yuanqi Du, Charles Harris, Arian Jamasb, Igor Iashchishyn, Vladimir Borislavov, Hannes Stärk, Joseph Lam, and Bruno Correia. Structure-based drug design with equivariant diffusion models. *Nature Machine Intelligence*, 6:1257–1268, 2024.
- [11] Jiaqi Guan, Xiang Zhou, Jian Peng, and Jianzhu Ma. DecompDiff: Diffusion models with decomposed protein–ligand priors for structure-based drug design. In *International Conference on Machine Learning (ICML)*, 2023.
- [12] DrugFlow Contributors. DrugFlow: Structure-based drug design with flow matching. *arXiv preprint arXiv:2407.08870*, 2024.
- [13] Zaixi Zhang, Qi Liu, Zhaocheng Wang, and Yang Li. Full-atom ligand generation with SE(3)-equivariant flow matching. *arXiv preprint arXiv:2407.08870*, 2024.
- [14] Helen M. Berman, John Westbrook, Zukang Feng, Gary Gilliland, T. N. Bhat, Helge Weissig, Ilya N. Shindyalov, and Philip E. Bourne. The Protein Data Bank. *Nucleic Acids Research*, 28(1):235–242, 2000.
- [15] ESField Contributors. ESField: Physical opportunity site utilization as a missing dimension in structure-based 3D molecular generation. *Technical Report*, 2026.

- [16] ESField Contributors. ESField v6-D.2: Analytic field-based guidance for HEW site displacement—a NO-GO diagnosis. *Technical Report*, 2026.
- [17] Christoph Griesbacher, Andreas Habring, Lukas Bogensperger, and Thomas Pock. Generating physically consistent molecules with energy-based models. *arXiv preprint arXiv:2605.18381*, 2026.
- [18] Zhan Li, Han Hu, Shao Heng Lim, Xiaodong Li, Fei Gao, Enmao Diao, Ziming Ding, Michalis Vazirgiannis, and Henrik Boström. A kinetic-energy perspective of flow matching. *arXiv preprint arXiv:2602.07928*, 2026.
- [19] Wengong Jin, Regina Barzilay, and Tommi Jaakkola. Junction tree variational autoencoder for molecular graph generation. *International Conference on Machine Learning (ICML)*, 2018.
- [20] Prafulla Dhariwal and Alex Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [21] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2021.
- [22] Arpit Bansal, Hong-Min Chu, Avi Schwarzschild, Soumyadip Sengupta, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Universal guidance for diffusion models. In *ICLR Workshop on Generative Modeling*, 2023.
- [23] Huafeng Lai, Tianhua Wang, Hanane Sirelkhatim, James Eaton, Hua Huang, Brad Rees, Ola Engkvist, Jon Paul Janet, Xiaoting Wang, and Alessandro Tibo. Improving protein–ligand complex generation with force field guidance. In *NeurIPS Workshop on SimBioChem*, 2025.
- [24] H. Y. Isaac Lam, Samuel P. Ojeda, Michaela Brezinova, Johannes Hanke, X. E. Ong, Yuguang Mu, and Michele Vendruscolo. Metadiffusion: Inference-time meta-energy biasing of biomolecular diffusion models. *bioRxiv*, 2026.
- [25] Lingle Wang, Bruce J. Berne, and Richard A. Friesner. Ligand binding to protein-binding pockets with wet and dry regions. *Proceedings of the National Academy of Sciences USA*, 108(4):1326–1330, 2011.
- [26] Thomas A. Halgren. Merck Molecular Force Field. I. Basis, form, scope, parameterization, and performance of MMFF94. *Journal of Computational Chemistry*, 17(5-6):490–519, 1996.
- [27] Anthony K. Rappe, Carla J. Casewit, K. S. Colwell, William A. Goddard, and W. M. Skiff. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *Journal of the American Chemical Society*, 114(25):10024–10035, 1992.
- [28] Oleg Trott and Arthur J. Olson. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, 2010.
- [29] G. Richard Bickerton, Gaia V. Paolini, Jérémy Besnard, Sorel Muresan, and Andrew L. Hopkins. Quantifying the chemical beauty of drugs. *Nature Chemistry*, 4:90–98, 2012.
- [30] Dan Friedman and Adji Bousso Dieng. The Vendi Score: A diversity evaluation metric for machine learning. *arXiv preprint arXiv:2210.02410*, 2023.
- [31] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.
- [32] Norman Cliff. Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological Bulletin*, 114(3):494–509, 1993.
- [33] Jeanine Romano, Jeffrey D. Kromrey, Jesse Coraggio, and Jeff Skowronek. Appropriate statistics for ordinal level data: Should we really be using t -test and Cohen’s d for evaluating group differences? In *Annual Meeting of the Florida Association of Institutional Research*, 2006.